AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM311
Information Security

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 15/April/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60 marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and** ID on the External Answer booklet.
- **Calculators** are **Not** allowed.

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PART 1: Multiple Choice Questions [10 Marks]

Questions 1 [10 marks] (1 mark/each):

1. **Which of the following best describes cybersecurity?**
   A. Only encryption of data.
   B. Protection of physical facilities
   C. Collection of tools, policies, safeguards, and technologies to protect cyberspace
   D. Protection of networks only

2. **A security attack is defined as:**
   A. Any accidental system failure
   B. Any action that compromises security of information
   C. Only malware infection
   D. Unauthorized physical access only

3. **Passive attacks primarily involve:**
   A. Data modification
   B. Service disruption
   C. Eavesdropping on communications
   D. System shutdown

4. **Privacy by Design (PbD) requires privacy considerations to be included:**
   A. After deployment
   B. Only during testing
   C. At every stage of system development
   D. Only during maintenance

5. **Sensitive PII includes:**
   A. Work email address
   B. Public phone number
   C. Medical information
   D. Job title

6. **Anonymization differs from de-identification because:**
   A. It allows re-identification easily
   B. Data cannot be linked back to individuals
   C. It requires encryption
   D. It applies only to financial data

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7. **The National Vulnerability Database (NVD) contains:**
   A. User passwords
   B. Software source code
   C. Known technical vulnerabilities
   D. Personal information

8. **Advances in technology have affected privacy by:**
   A. Eliminating privacy risks
   B. Increasing ability to protect privacy only
   C. Both improving protection and compromising privacy
   D. Having no impact

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9. **The principle of least privilege means:**
   A. Users receive maximum access
   B. Users receive minimum access needed for their job
   C. All users share equal access
   D. Only administrators access systems

10. **A privacy risk assessment primarily helps organizations to:**
   A. Eliminate all risks
   B. Determine appropriate privacy budget and controls
   C. Avoid legal requirements
   D. Increase data collection

PART 2: Short Answer Questions [25 Marks]

P2-Q1 [5 marks]: Explain the concept of cybersecurity and describe the types of assets it aims to protect.

P2-Q2 [5 marks]: Define privacy and explain how privacy by design differs from privacy engineering.

P2-Q3 [5 marks]: Compare de-identification and anonymization of data, including their purposes and limitations.

P2-Q4 [5 marks]: Explain the role of the National Vulnerability Database (NVD).

P2-Q5 [5 marks]: Explain the principle of least privilege and why it is important in access control.

PART 3: Problem Solving and Case Studies [25 Marks]

P3-Q1 [15 marks]: Study the following cases and answer the related questions:

CASE-1 [3 marks]: **"A hospital stores patient data electronically. An attacker gains access using stolen credentials and modifies medical records"**. Suggest two security mechanisms to prevent such attacks in the future.

CASE-2 [3 marks]: **"A social media app collects extensive user data and only adds privacy controls after complaints"**. What principle of privacy is violated?

CASE-3 [3 marks]: **"An organization stores employee names, work emails, salaries, medical records, and bank account numbers"**. Identify which data elements are sensitive PII.

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CASE-4 [3 marks]: **"A fitness tracker company collects user location data for navigation features but later sells the data to advertisers"**. Identify the privacy threat involved.

CASE-5 [3 marks]: **"A software vulnerability allows attackers to remotely access a system without authentication"**. Using CVSS concepts, explain why this vulnerability is critical.

P3-Q2 [10 marks]: A university information system manages student records. The system has the following subjects and objects. Construct the Access Control Matrix for this system.

Subjects (Users)
- Registrar
- Instructor
- Student

Objects (Resources)
- Student Database
- Course Grades File
- System Configuration

Access Rights
- Read (R)
- Write (W)
- Execute (X)
- Delete (D)

The security policy specifies:
- Registrar: Full access to all resources
- Instructor: Read/Write access to Course Grades only
- Student: Read-only access to Course Grades
- Only Registrar can modify System Configuration
- No user except Registrar can access the Student Database

End of questions

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